Focus group meeting - 1st Group-20250604_110536-Registrazione della riunione

4 giugno 2025, 09:05AM

1 h 31 m 38 s

Autor1 0:05

OK.

As I was saying, thanks everybody for be here and.

To participate in this this focus group.

Basically, we will have first first part in which will we will introduce the problem and the challenges to have also a big background on the on our approach and then we will then go.

Go to technical details with Pre recorded demo.

I will share my screen.

OK.

So I think you can see my screen.

Author 2 0:59

Yes.

Author 1 0:59

OK, OK.

So.

OK, as I was saying.

1st a little bit of introduction on on the Focus group.

Basically, focus group are strategy to to perform qualitative assessment.

In in for, for research and in particular to evaluate the quality aspect.

On on on approach.

Or in our case, basically they follow the structured form that we will follow during this interview.

Based on top of.

Direct question about both of the expertise of each participant and.

The the technical details of the tool basically.

Adjust the the participants.

So that.

That participate. Basically we have we have different backgrounds and affiliation.
So this is good.
I mean for.
I mean having a good feedback on our tools, our tool and of course.
Having different.
Backgrounds and.
And opinions?
So as I was saying, we have two parts. Basically. The first part is background introduction on the on the topic.
That is, in particular, the analysis of failure propagation in IoT system.
And we basically.
We were also open challenges and limitation of existing approaches.
That motivate our solution basically.
So.
Just for.
End of day.
For, for letting you, letting you know you can you.
Can you can intervene?
Whatever you want, even though we we planned some some some time for dedicated discussion after.
The the 1st and the 2nd part let's say, but if you have of course any any question, you can raise your your hand and or just turn on the microphone and.
We will.
We will answer OK.
So let's start for the background part basically. So in IoT system.
A lot of research has been has been done for failure propagation.
From analysis and in particular because those system are particularly prone to different kind of failure.
That of course depends on many factors, in particular on the environment.
And due to this variability.
They their analysis should be could be complicated or difficult to conduct, of course, because we have different system and also different.
Components that.
Can be, can be, can play a role in the overall system.
And this is just a motivating scenario that we we will will bring through through all the discussion.

Basically, let's let's think about smart smart irrigation scenario. That is the typical IoT.
The typical involves Internet of Things components.
So basically we have.
We have different different components as you can see.
In the picture, so we have for instance a sensor for temperature or humidity and then we have of course the lens to trigger some action.
According to of course, the loop that is typical.
The the methodology that developers use to to implement IoT system and of course we have also the.
The web parts.
So the the the Internet partners. So with with our cloud server for instance, that collects the data.
Retrieved from sensor and on top of this data. Of course the system.
Can can can decide who can can act.
In autonomous in an autonomous manner, basically.
So the problem here, as you can see is that when Nafi elura happens.
Umm, uh. We need to locate this fellow according because because the the propagation of this failure may affect all the system.
In for for particular, for particular in particular situation, for instance.
A failure in the in in the sense or.
Water loss or an incorrect operation, for instance.
The the system can decide.
And the two to irrigate the plants when but but actually is not required because of the field.
So this this of course can.
Can have different risk.
And also.
On the on the storage, for instance of the of the data.
And of course, the type of failure.
Is of course depends on the time of the of the.
On the domain and in particular the system requirement and the specification.
An approach that has been developed is the so-called logic analysis in which basically we have a formal set of rules that.
Is capable of.
Of representing the behavior that may happen in a formal.
With a formal notation.

And basically these rules are are defined by the user or are defined by the user and and in particular observing.
The the behavior of the system.
And the also an approach to do this is to introducing.
Let's say.
Synthetic data or failure.
That that, that can trigger the fellow process and so can can help the the user to understand.
The the the logic basically. However, as you you may you as you see this this process it's errone and of course requires effort.
And effort, Manuel.
Effort from delivered from delivered speaker.
This failure you're you basically the developer have to run the system to to first of all identify those failure and then propose there was that can handle this failure can predict this failure in advance.
So the main objective of our work is automating as much as possible.
The the failure profileation discovery.
Based on top of these rules.
And of course.
Another problem that has been not addressed by is.
Giving a comprehensive environment to do so. Let's say a controlled environment also aligning on simulation, so.
And in particular, for this demo, we implemented our system on top of protus. It is dedicated to simulation.
For.
IoT components basically.
Ian, just to present better the what type of failure?
Can be identified in IoT system in our in our peers work we we identify this set of failure that you you can see basically.
We we also are are assuming.
That the system the system is a time time based system. Thanks.
Which is mean that the system expect the time series or.
Expected basically.
A value in a certain a certain time window in particular. So as you can see the failure are built. Let's say have been identified on top of this.
Definition. So we can have, for instance, if the the system provide.

Avail too early. We can have an early favor.
An early favor type.
Let's say so in this case this can cause different failures to other to other parts of the system or other components.
According of course to the specification.
For also another type of failure could be omission.
So when Davilu expected is not provided by the system, this may cause as I mentioned before.
An unwanted the behavior from the system, for instance.
This may trigger the irrigation in our smart irrigation plant.
Of course, as Lindi also we have a problem in a sense that this set of failure may not.
Exhaustive according to of course the the type of IoT system. So what?
What we want to obtain is a system that is genetic enough to allow developers to specify their own rules for failure and test the the system in control and environment.
Before, of course, deploying the system in a real world using a real world or real IoT components.
Yeah, as a as a as I said before, so also.
Existing approach. There are the existing approach that relies on, of course different different approach, different techniques or automation but.
We hope that.
That most of the approaches do not support time based system that actually are quite spread in the real world application and.
Supporting the different simulation platform is still an open challenges. While we we focus in this demo with without a single simulation platform, that is produce, we also will. We will also show that our system is flexible.
To include additional simulation platform and of course.
The.
Most challenging part is to formalizing the rules that helps the developers to analyze the system.
So of course this this tool, our tool will we will in that just OK just to.
Have a snapshot of.
A traditional execution without our our DSL.
This is a bit spoiler because we we will go using ADSL approach domain specific language approach. Basically what?
The.
The developers have to do is run the system.

In this case, using a simulation platform.
Of course, all these tests that should be in should be run in considering an isolated component. In that case we will show that.
It's our.
System, of course, without the web apart. So we are considering basically only the irrigation and sensor.
Part.
For this, for this demo. As you can see then.
They they sort of the component that is fed with an input time series or a list of time series and after run the simulation we obtain the the output. Of course the output time series because the system is defined as a time based IoT component.
And in the meanwhile, they the user, after running this at two specify the manual, the manually the the rules.
Of failure propagation. Of course, this, this, this process can be seen as an integrity process which for instance the developers have to run multiple time the system to to have to have an intuition of the definition, or alternatively the user can.
Define can rely on previously defined rules.
In the end.
There is a this.
The final step is failure.
Discovery that is basically.
The application rule failure of the rules to the obtained the time series.
So it's ANC if the failure is happening somewhere in the system in this in that in that case the isolated component help us to.
To have a good estimation, or at least to isolate the failure, I mean where the failure is is happening. As you can see a lot of manual step we are we have a lot of manual step in this process.
OK.
So this concludes basically the first part.
So.
So here we you can.
You can briefly introduce yourself to the others.
And say thank in very few words in your experience.
And of course, if you have comments on this first part, please, everything is clear.
You can.
You can of course.

Ask us.
Me. OK.
So.
Who wants to start? Maybe.

Author 2 17:34
Maybe we can follow the order of the first slide just to have a guideline, sure.

Author 1 17:37
OK, so I will go back.
So OK.

Participant 1 17:42
Play.

Author 1 17:44
So **omissis**.

Participant 1 17:45
Repl.
Yeah. Good morning, everybody.
And yeah, can you bring back the questions so that I can follow the?

Author 1 17:54
Yes.

Participant 1 17:57
Demand.
OK.
So my experience with modeling language tools OK, I'm quite wide knowledge in using them, in particular in the area of software.
So UML or this kind of modeling languages, but also I work with.
Some modeling.
Languages for your tea systems.
That that, I mean this kind of flu based modeling tools?
I have some experience with Diot.

Not much with simulation, but with the framework for IoT platforms like, I mean that you can also simulate something like things border, stuff like this.
So IoT platforms that includes simulation components somehow.
And I mean my research activity is mainly in assurance of complex software system. I mainly worked on software testing for a long time.
A different level of not much in the area of IoT, let's say where I did something different.
But yeah, in general in terms of web services or now I'm working on blockchain, for instance smart contracts, so I.
Have dynamic verification strategies.
Somehow are.
The topic that I investigated the most during my career.
So this is somehow the profile and now I could contribute to with my competence.
I think this group, so I I think I have to keep it short.
I mean it's.
I think it's enough for what you need.
Otherwise, ask.

Author 1 19:43
No, no, thanks. Thanks. He's perfect.

Participant 2 19:46
I think that I am the next one, **omissis**.

Author 1 19:49
Yes.

Participant 2 19:52
Hello everyone.
omissis Quality assurance director for ADO for who?
Don't know what the other is.
Other is a sport technology company that provides video and data analysis platform for coaches and that lead at rates to improve their performance in this space.
I'm also a contract professor for the Computer Science course in the University of.
Genoa and I don't have experience in modeling language tools or some academic experience, but nothing of relevant for this focus focus group.
And also with IoT, but I'm think an expert on software quality. I started work in this area.

15 years ago, 20 years ago.
I don't remember exactly.
And this is my main role in my current in my current company.
So this is this is my quick introduction. Thank you.

Author 1 20:46
Thanks **omissis**..
The next is **omissis**..

DO **Participant 3** 20:51
Hello everyone.
I'm **omissis**. and I'm the postdoctoral researcher for the University of Genoa, and my experience with modeling language tools is mostly UML, especially class diagrams and sequence diagrams and state machine diagrams.
My experience with IoT simulation platforms in during my master thesis.
I wrote.
A simulator for.
For an smart insulin pump that.
Reads that reads the glycemic value from the patient, sends it to a cloud system that analyze it, and then the cloud system commands back to the smartphone and then to the pump.
What action should be done?
But it was all simulated using nodrad Android emulator without any real hardware.
And this is all my experience in IO TCM simulation in evaluating software quality.
My field of research is mostly web testing, so it's this is my my experience in evaluating software quality is mostly end to end testing of.
Applications.

Author 1 22:25
OK.
Many thanks **omissis**.

J **Participant 4** 22:31
That's right. My experience with modeling language tools I've been developing modeling language tools with visual notations based on X and also the textural representations. Now the last more than seven years with multicore.

So we created such modeling language tools ourselves, and also the adventures to create. The tools.
My experience with RT simulation platforms is rather low.
I of course supervise the feesis which was developing modeling language and tool platform for developing IoT systems.
But simulation was not so much in focus of that one.
So we have some simple step by step simulations, but not very complex ones.
And we developing software quality.
I did some software architecture reviews for industry.
The companies and also a little bit of model based testing, but I would say my most experience is in #1.

Author 1 23:40
Thanks.
omissis. is next.

Participant 5 23:46
Hi. Hello everyone.
I am currently a postdoc in Switzerland.
So about the first point, I would say also in my case I academic experience here, but it's a bit trusty because it dates back to my university years.
Regarding IoT simulation platforms, I happen to read some papers.
Testing, but yeah, not so much practical experience.
We had a grass, so when it comes to reservation and yeah, I would say that I have quite some experience in software testing.
Especially in test generation and yes, both in the system level and in the unit level.
That that's not as it. Thank you.

Author 1 24:38
Thanks. Thanks to you Pablo to conclude.

Participant 6 24:44
Yes, thank you, **omissis**.and everybody.
Well, I'm for those of you who don't know me yet. I'm a member of the Michel team in Madrid.
Since 2015.

And while.
I.
I work.
I work there creating some Dsls for.
Model mutation and also for.
A model based metamorphic testing. These Dsls are mainly textbook.
So I think I I have.
A high expertise with Dsls design and creation.
Regarding the second question.
Well, I I haven't worked directly with IoT system.
So I think I have a low expertise in this regard.
Some colleagues work with.
I think they call it fog simulators.
It's I think it's like a simulation of these IoT systems.
And I've worked with cloud system simulators.
But in in the in the indirectly just applying metamorphic testing to to them with one of our metamor based metamorphic testing framework.
And about the last question.
Well, I would say that that I have an average expertise.
Or a little bit higher than average, but from the point of view of software testing.
I've been teaching this course.
A course on software testing and well, most of the DSL SI I create.
And design and this model based tools.
Are.
To apply software testing, invitation testing and autonomous testing.
So that's my expertise.
I think.

Author 1 27:17
Thanks. OK, so.

Author 3 27:25
OK.

Author 1 27:27
Yeah, I I can leave the the floor to you ready rockco?

Author 3 27:30
Yes.

Author 1 27:30
That is the main implementer of our tool and also.
It will show you the.
The tool itself.
Yuri, I think I think you can take control of my screen. Yes please.

Author 3 27:46
Yes, I can take a look. I guess that I'm sharing my screen.

Author 1 27:50
I I. Oh, OK, OK, OK.
Don't worry, don't worry.
So let's go ahead in this way.
Yeah.

Author 3 27:57
Can you see my screen?
OK, perfect.

Author 1 28:00
Yes.

Author 3 28:01
So here we are.
We are describing how we implemented our tool and we will support the process of failure discovery with the DSL. So in particular we are defining a DSL for defining components that has for input and output ports, and then we have injection rules as well as discovery operation that.
Is able to take the outputs and then try to infer the possibility to define.
To identify failure risk code. In particular, we have to define the DSL and the concrete syntax we implemented by means of extracts that provide a lot of features.
For managing with the syntax and then also with the semantics.

So we define with rdsl.
We based it on definition of components. As I said before, components can have input and output ports.
And then, without DSL, we can define injection rules.
Injection rule is the way that we can modify the time series.
Time series are pairs of series of pairs, time and values.
We can define some injection order in order to modify these time series that are in the input of the the system and in the same way we can define.
Discovery rules that is able to analyze the output time series.
And this color potential potential failure for sure in order to support the user with different.
Operation we use.
We define it as under library. They provide a lot of facility that allow the developers to easily implements both injection rule and discovery discovery functions.
So just to show you how that the process have been changed by using our approach, here we have the user that defines.
And without DSL, how the system is composed by means of components, they're they're both.
The injections and the injection and the discovery rules and the system is able to use the standard library to reuse facilities and implementing those rules. So once we have these rules, the input time series will be mutated by the injection rules.
Then we submitted to the simulation part the submit the submission part as some output.
Output series that are modified.
And then Discovery will check both the original output with the ones that have been obtained by simulating the system with the mutated injection rules, and then they find the discovery part.
So at the end the user should just provide the pairs of injection rules and then discovery functions.
So by back to the delegation delegation scenario that Claudio presented in the in the previous part, we have two components.
Sier the components are highlighted with the rounded boxes, and in particular we have two output parts that enable the the plants and two input spots that give the the the pars of humidity and temperature.
So just to to be concrete here we can define a file that is an embed files these emilop files.
When you start the plugins in the eclipse, all the infrastructure plugins and we have a well documented supporting repository.
Sky Boutique start the plug insurance so so we can define EM a lot. Files that are the

specification that includes components injection and discovery rules and then we will be we'll immediately generate the Java files that is able to run the word platform including the mutation the SIM.
And then the discovery part, so these all these these parts will be showing via concrete example that Claudio we will play after.
These slides.
Carl, you can you.

Author 1 32:05
Yes. So I share again the screen.

Author 3 32:08
Yes, thank you.

Author 1 32:10
'Cause we have the video.
OK.
OK.
This is the demo.

Author 3 32:20
What?

Author 1 32:23
OK.
So this is basically the process overview, the simulation platform suite that we we are using for implementing for validating our ADSL. Basically as you can see here we defined.
To import parts and the outputs as Yuri was saying before, so the component and of course those are the elements that.
Produce user to measure the final results.
So, OK, coming back to the Eclipse project, our our tool works as a plug in process.
So plug in projects so you can import in your.
Basically, in your eclipse environment.
We also we we have dedicated gitan prepository that helps the developers to of course install the the tool and.
As you can see, set all the required dependencies and one of the dependencies of course is

the DSL and the standard library that Yuri was mentioning to help.
Of course, the user for the Java part.
So OK.
Once you set up the the project, we can specify our our DSL file. In that case we call it the irrigation unit.
And of course, the main technology for those that are experts on this technology are X text technology that allows you to allows development to specify DSL specifically. As you can see the developer sees is guided, the developer is guided.
With basically extracts the facilities, so syntax syntax highlighting.
The auto completion parter and of course if there are errors in the specification as in this case.
So in that way we have a full full full support during also the specification.
So we can say that the specification needs semi automated.
Here we basically in our experiment we encoded the time series as text text files in which of course.
The user can specify the tuple of time and values.
Of course.
Then that will is extensible enough to ingest the other type of specification.
Like for instance Jason file.
Or other kind of structured file moving to the injection phase. As you can see they use.
This is the same file.
You know the same specification file during the specification of the the file they use on the standard library that we manually developed.
In that case, basically there are specific Java function.
To support.
They of course the injection of veggies.
In that case, we are simply adding.
A time value.
Adding both time and value to the time CS. Of course, expert developers can also rely on Java traditional.
Structure and classes to define their own injection.
On the other way around, we will we will also facilitate the non expert using by by by for instance introducing a sequence.
Custom failure and for instance omission.
In that case.
Here also, this is the discovery part.

So that the last part of the Dula, as you can see the.
Developers can.
Sorry there is.
Some.
OK.
Yeah, we OK.
Just to wrap up here, the let me pause a bit just to explain you.
Year 20 user again to have a structure.
To have a structured let's say DSL.
Here we we also put a config concept in which the developer can map the injection to a specific port.
And of course coordinate configure the discovery phase that we just saw.

Author 3 37:30
If I can add this part is able to make the mutation rules and discovery rules agnostic from the the time series. It means that we can reuse for different components and different parts. These injection injection rules and discovery rules.

Author 1 37:50
Exactly. So these are also more flexibility as you can see then of course.

Author 3 37:56
Yeah.

Author 1 38:00
The.
The developers can go back to the specification and and we have also javadoc that is automatically.
Generated as you can see here, the the simple injection that I was showing before.
And here.
Can also map the discovery to for instance. Here we want to to discover the omission, the mission failure as you, as you may remember, is when.
The the.
The venue is not provided, so there is an omission in the expected value.
Yeah, of course.
The the the the can navigate the the the the specification as.

Author 3 38:57

As you can see, the omission should be taken consideration the port and then the expected time series and the ones that is obtained by the simulation.

We did injected series so here you can see that is related to a port.

There is an unexpected time.

Expected time series? Yes, exactly.

And we define edit in the configuration in the top of the file and then also.

The ones that these are computed by the the automatically computed by the simulation.

Yeah, Jeff, I'll see if exactly nuance that has been obtained by disimbigration.

Author 1 39:37

Yeah, they injected the CDs and it's in this case.

Author 3 40:05

OK.

Yeah, another discovery and discovery function.

Discovery configuration where we are checking the the custom failure. As you can see the custom failure rely on out time series and expected time series that are the ones that we are configuring by the configuration part.

Yeah. So now we are writing the port where we would like to check.

Expected series and the injected series as well.

Once we completed all the specification, this is a a very small one.

Automatically we generated a Java file that is able to run the whole platform, including the simulation part.

Author 1 41:15

Yeah, yeah.

For instance, there is the the simulation part.

Basically, we developed a set of function that simply call produce our environment and do the simulation.

Of course, this part can be extended and the other this is the the PROTOT project with the inputs output expected from the simulation. Of course here.

We can.

Er, you can. You can plug different simulation platform, but we have to extend our of course internal library and this is as Yuri was mentioning, this is the running basically.

Author 3 42:09

Oh that the jar file has been generated by the syntax.

We can run the whole platform that automatically injected the time value to the input time series and then simulate the platform. As you can see here. So that is running the simulation platform that is produce in this case the system is running and obtaining the value for inst.

Here you can see and then save the output of the value after the simulation.

Once we are painted to define the output files, the discovery part will be triggered and then automatically we can see the TR. There is a custom failure report issue because we have the differences between the the expected value and the time value as and another result that will.

Be show and we in the Jason files where there is.

Error and the reasoning of the error.

OK.

Author 1 43:12

OK.

Author 3 43:14

Can you present last three slides?

Yes, Sir.

So yes, this one 2323.

Author 1 43:22

And yeah, from the standard memory.

Author 3 43:25

OK.

So the standard Library is a component that provides facility.

In particular, there are different facility for supporting injection discovery and also utility operators that allow you to play with the type series.

In particular, the standard library has been extended by the producer services because, as you can see the the the word platform is able to simulate.

The the the input and the execution process, but the protocols platform here we will be agnostic by the protocol service, but for sure for running this example we we create a

project services that extend just the the part of the simulation and the input output.
Input part to ending karlio.
Can you move to the next slide?
So.
Yes, but what we would like to do that by extending the standard library by different elements, for instance, not red.
Or component we can extend our tools to the fit DSL to support different simulation platform.
Now we are at the end of the presentation and we left you with some questions, in particular, in order to comment on the useful of the proposed DSL.
Guiding specification execution or foil or analysis and.
The second question that we would like to know your your your answer is the comment about experience is preveness of the DSL with respect of mainframe is over federal analysis.
And finally, we would like to know some feedback for you about how to improve our text and implemented tool by means of different different announcement or modification.
So I guess that we can use the same the same loop that we used in the first in the first round of questions.
So when the first target was.
Andrea yes.

Participant 1 45:38
OK then I am the one starting with a question.
No, we don't answer.
No, just a clarification.
Yeah, because I would like to better understand.
I mean to clarify some aspect because maybe I had also some problem with connection sometimes or maybe you say something that I didn't catch it correctly. What I would like to understand better is OK you somehow you have time series somewhere somehow that represent real value real real.
Data.
That you observe them on real environment, I guess.
OK.
This is true.
Can you confirm this? So you start from these assumptions, OK.

Author 1 46:15

Yes.

Participant 1 46:18

And then you the the kind of.

Errors that you would expect are just the one that you were mentioning a single output.

Somehow you do not expect.

These, I mean it's delayed.

It's, I mean, the kind of fault you are trying to catch of that one.

So my first question is.

Because I imagine that there are other kind of errors that may be in time series you would like to.

I mean, you can express with logic and stuff like this, obviously kind of temporal logics and then you spot this because the because what I don't see somehow.

But I don't know the status of the art in this domain.

The first question I have is why you do not derive the injection from the. What you would like to get to catch.

So why you do?

Why is not possible?

Because I show that you manually insert.

A value that is a kind of mutation that you would like to have and my question is why are not there any technique that you could insert?

Because this goes to the first question.

Usefulness of the proposal DSL regarding the specification. What I mean is obviously it's useful, but I think it's not easy for it's my impression.

For someone that would like to mutate the time series to identify how it should mutate it in a clever way.

If I got it correctly.

For me it's easier instead to specify what I would like.

What are the rules that I think should be?

I mean the kind of assertion that I expect to be true and then from this assertion to derive kind of mutation.

But I'm not sure it's. I got it correct and so on and so forth. So this comment on usefulness, I think it's useful.

But what?

I think it's that.
At the same time that it's not easy for someone to understand where to put this mutation.
Which kind of mutation?
Which is the best one?
Because at the end you have.
You don't have much time, so probably maybe in this election it's not really driven by. If I got it correctly again.
So everything is good. I mean it's. But if I got it correctly, I mean, you know, I I don't have much time and I need to.
Go to the best.
Mute.
Somehow I'm I'm not really guided.
For what I got, at least if I got it correctly.

Author 3 49:13
Andrea, if I got your your question.

Participant 1 49:17
Yeah.

Author 3 49:17
I guess that you would like to understand depending on the failure which are the the mutation that are being applied on the on the input time series, right?
So the other way around.

Participant 1 49:30
I mean you you apply the invitation.
You you are the the one applying the mutation right is the programmer is the I mean the and so my but at the same time you are also the one that would like to specify this kind of assertions. What?

Author 3 49:36
OK.
Yes, accepted.

Participant 1 49:49

Is what do you want to check on the time series that is correct or not?
Is it true?
I mean you.
You specify both the mutation and the assertion.
That you you you see.
You think you should be somehow verified on the on the time series?
So what I've my idea is that somehow you could derive the mutation from the assertion and so that can be selected. The best one could be selected according because they're fine.
The mutation is somehow a complex activity. If you are not guided, at least this is for what I understood.
And I mean, this goes also back to the expressiveness. I mean, I think it's, it's OK.
I don't know if this kind of you can enlarge the kind of.
Checking that you would like to to to, but again I'm not.

Author 1 50:51
Yeah, yeah, maybe if I probably got your point.

Participant 1 50:51
I'm not.

Author 1 50:55
Of course it will.
Is of course the the target user is of course someone that at least is expert and wants to.
So let's say you know what?
I bought this is the developer at least knows what kind of?
Strategy, or at least some.
Some uh, intuition of the the kind of injection, but I agree with your point.
And in that case, we can improve the documentation of the tool. Basically on the GitHub repository in which we we provide we provide.
A set of guidelines and of course.
More technical overview on the OR on the type of injection function that that is available on the tool.
In in particular, I asked that I show it in the video.
We we already we we already did some.
Some predefined if we want to call it function in the standard library that the user can use

can can directly call without specifying any parameter, because for instance if the user wants to implement an onision, there is already a function that will do it. But of course I. Agree that not all the user.
Understand when there is needed an omission to get a particular failure in that case. In that case, in this case we we can improve the DSL by providing.
A real time documentation, or at least.
An explanatory example on how to use it, but indeed these are good fit, yeah.

Author 2 52:48
But I would like also to to add some comment on this because Andrea if I got your your point and maybe you're, I mean the text testing experts here can can help me on this. So essentially.
You're suggesting to have something similar to what you have with the testing.
You know, so supporting the generation of testing or a way to, you know, to to get the testing.
Generated in a clever manner starting from.
The source code of your system, for instance, right?
So in this case, what do we propose?
Here is machinery is, you know is.
The Maining, the main building block in the ingredients to support this process. The process of injecting something to your component that is typically black box so you don't know actually what what's the behavior of this component. So you do injection of time series and you according to what?
What you know about the component?
Know what should be the expected outcome and then you compare what has been obtained with the expected one and typically all of this is done.
In.
Let's say without any specific support and what we propose is a machinery is a support to, you know to to put this process in place now.
We did.
And of course you you can use.
This machinery, these main building blocks. If you want to propose a clever manner to write the input, you know to inject the injections. Of course you can.
You can do that right?
But again, this is the machinery, as you know, as a building blocks and the way you generated the injections is is up to you.

Participant 1 54:39
Turn off.

Author 2 54:43
Now it's kind of.

Participant 1 54:43
Yeah, yeah, I can go to the.
You can add the layer on top of it with the strategy.

Author 2 54:45
No.
Exactly. Exactly.
That's the point.

Participant 1 54:49
Yeah. Yeah, no, I mean, and I would like.
I think it's useful that also the others so, because otherwise maybe I misunderstood something.
So for your purpose, I think it's important to get many other.
So now it's a bit more clear understander.
So you want to build a basic layer in which someone can build on top. OK, yeah, OK.
So I think it's useful if the others can say the others and then we we recap.

Author 2 55:15
OK.

Participant 1 55:15
So thank you, **omissis**

Author 2 55:15
Thank you. Thank you, **omissis**

Author 1 55:21
Sorry, go back to the **omissis**. Yeah, OK.

Participant 2 55:23

I think that I'm the next one, yes.

Yeah, so sorry.

Author 1 55:27

Please.

Participant 2 55:27

Maybe I am really busy question, but who is the target user for this DSL?

So I'm really not an expert.

So maybe in the IoT developers and reliability expert and I wonder if you have validated the the usability of the language with them.

Author 1 55:48

You mean with users?

Participant 2 55:49

Yes.

Author 1 55:50

No, no, no, actually no.

And because we we won't have to. This focus group actually is a preliminary step, let's say, to do to do so.

But concerning the target user, I think as I was saying before.

The user should.

The developer should be knowledgeable, knowledgeable enough of course of.

Testing or failure rules in that particular.

But not so much about simulation and.

Because because basically the the process is basically automated, but indeed.

A user can also rely on an expert domain for, for instance, for guiding the the discovery phase, even though it's supported by.

Uh by by the DSL basically and and of course.

A basic knowledge of domain specific languages required even though we tried to have very, very generic concepts.

So the the idea. Ideally the user plugged the tool specifying the the.

Specification and then run the tool.
But of course.
Yeah, should be somehow expect.
Or at least.
Have knowledge about.
All time series basically works.

Author 2 57:30
So if if I got if I get complimented, this is essentially the same, OK.

Author 1 57:31
The basic concept, yeah.

Author 2 57:37
Short short answer is a software developer.
So the same kind of people that are using the prototype and they want to complement the simulation protocols with this additional functionality. Concerning the the failure analysis.
So but this software developers at the end.

Participant 2 57:54
OK.
Thank you.
And maybe maybe another additional silly question, but so how do you plan to manage the validation of the model against the utter behavior of the IoT system?
So usually so my understanding is that the the IoT system are complex and really dynamic.
I wonder if there is any way to validate.

Author 1 58:19
In our previous work, we basically manuelate evaluate manually in the sense that.
Relying relying on the the closest set of failures. For instance, Mission Commission or early failure, we basically run first the devolution manually and then we compare the with the expected result.
So in terms of, yeah, in terms of DSL, we we plan to do.
At least this part, in a sense, even though we can apply some automated metrics.
For instance, given given the expected results, we can say if the the injection worked in a sense, modify.

Precise value, or at least and and of course the discovery part can help in a sense that we we also device a component that I mean the the concept of discovery.
Include the fact that we we have the the report, the final report, so.
It's like.
We we can, we can provide the different use case scenario for instance.
I don't know.
We can we if we expected an omission failure, then we would both manual and the the DSL approach to do so.
And if the results are.
Are.
The same we can say that this work but but also it's like we want to provide as we we discussed the before we provide also.
Abola and automated approach to also speed the development of code because in case of the in in without DSL the the without the the this DSL the the user have to implement by themselves.
And self also the injection rules and this is could be.
Time consuming in a way.

Author 2 1:00:28
Again, to follow up on this, **omissis**, I guess that you are referring on cases where you have a model given specified by means of our DSL, that includes components that for instance, are not the.
I mean exactly those that are real, the known and do not reflect exactly the real the real system.
So I agree with this point, even though again, by defining to the previous comment that we did with Andrea, here we are, you know, refining, we are putting in place the ingredients.
So the approach now the way you get the specification can be even semi automated.
Now imagine for instance you have the real system and you retrieve a real set of components. The real set of inputs.
So essentially you have, you know, a starting point that is close enough to the real system now.
So in any case, we have the language that allows you to specify the the system.
You. So you have the language now, the way you get this initial set of components.
If you do this manually or you retrieve this by analyzing the real system and automatically retrieve not just the components.

Again, this can be an additional component, an additional machinery that can help. Not in in getting the the specification of the system.
I don't if I answer your your question.

Participant 2 1:01:48
No. Yes, yes, thank you.
OK.
Probably I have other question also but I want to allow other people to ask.
So thank you for the answer.

Author 3 1:01:58
Just to complain, to compliment the the answer, Claudio, if you go to slide 20.

Author 1 1:01:59
OK.
Yes.

Author 3 1:02:10
So here you can see that we can define components among the real system deplo.
Ys simulation platform and the way that we are defining the components can be manipulated.
Towards with the DSL, so we can define different components. But the way that we would like to set input ports and component and output ports within a components is defined within the DSL.

Author 2 1:02:38
Yeah, but that was the problem.
You're essentially, **omissis**was saying.
What if you specify components that are not in line with the real system, right?
So you needed to be sure that what your model is actually reflecting the real system and but again we we provide the language for doing so and the way you specify this initial model can be even simulated, right? So in order to ensure that the model is act.

Author 3 1:02:50
Yeah.
Yeah.

Author 2 1:03:04

Reflecting the system that you are in in your app.

OK.

Thank you. Who is next?

Author 1 1:03:14

omissis I guess.

Author 2 1:03:16

Yep, no.

Participant 3 1:03:21

Hey. Hey. I have a question about at the beginning of the presentation.

You listed the different type of wrong values that can be present in the time series. Yes, that.

Author 1 1:03:41

He's one.

Participant 3 1:03:43

Value value late value value, value, course are clear for what concerns value subtol for output.

In the range about.

How do you decide which output is in the range about?

Author 1 1:04:00

Hey. Hey. Yeah, this is.

I think it's.

On the real value that.

Expected from the system in a sense that for some, for some for some IoT system.

There is a, let's say, a target value.

That.

Maybe also could be an error?

An error in the format in which the value is provided because for instance, we can imagine that.

The the IoT component for some reason.
Send some special character or maybe.
A value that is not a number.
Due to the maybe the sensor. Uh.
The sensor error that of course in the case we we analyze it the basically 11 isolated component, but in the real system we we can expect that.
A failure in one. For instance, in the sensor can lead to also an error in the communication of that value.
So also erroneous seats.
It's in this sense.

Author 2 1:05:24
Sorry if I can come.
Yeah.

Author 4 1:05:27
There we OK. So I would like to follow up on this.
I think this is also also exactly the point of the DSL and and the tool that is, it's not known a priori exactly how a value course a value subtore is defined until you know the system what the system does, and so this is exactly the point of.
The DSL the user could use our functions to the function of the DSL to define its own concept of various opt.
Even a new type of failure that is not part of radio this list and then this definition is applied on the output to discover the the failures.
So I think the question goes directly to the motivation of the tool.
That is all. Those are all those the different kind of failures that a user may want to detect is not known at preordiant.
Then you need the domain knowledge to.
Implement this definitions and the DSL is somehow facilitating the process of defining what the different.
Failures that a user may want to injected or discovered.
Participant 3 1:06:33
OK.
OK, that's clear.

Author 1 1:06:39

I think we can go to me.
I go to the Westerns.
I think Participant 4is the next.

Participant 4 1:06:50
Yeah. Thank you.
Thank you very much for your interesting presentation.
So for the usefulness I think.
I mean, clearly you need some documentation and tutorials and other things to to help getting into the language, but what I've seen from the first steps now for me the majority looks intuitively understandable, so you've also added enough syntactic sugar to the DSL so that it's clear what.
The statements are.
So that's that's quite nice.
See what you might need to make clearer is what are the particular use cases that you want to support. You also have already the question with who are the user groups? But I think it's also what are the user scenarios.
So what is the concrete or several concrete usage scenarios when I want to use the DSL for which purpose?
Because that's not fully clear to me right now.
That also comes then to to aspects related to usability and expressiveness.
So for usability for example, I'm wondering if how, how usable the DSL is if you have a large amount of sensors. If you think about the system, I don't know 50-60 hundred 50 sensors.
Then it might be quite a high approach that I have to put into as a developer to define.
So first of all, doing the specification, which is already quite long and then specify each of the injection rules.
So I think that will become quite long.
So if this is one of the use cases that you have in mind, then you might have to think about maybe summarizing things or reformulating aspects.
That is help we can make it faster and.
I think also.
When it comes to this, this injection rules I might have sensors who have the same who are the same pipe, who might have the same failures.
So therefore I don't want to define.
And for every single sensor exactly the same injection, but maybe for several of them

together.
But it's already clear that it's the same ones.
And also I'm wondering if there are more complex injection patterns, so it's not just one failure in one zenzor, but more complex ones that you might want to specify then also in a in a quite short way I don't.
I'm not sure if there are such.
Patterns of failures.
But maybe that will also be a nice extension.
Maybe for for the library that it can be find the combination of these things. For example, for for particular aspects.
And.
The last aspect for the discovery, so currently we have not seen so much of what what should we do with the results now.
So I'm I'm not fully aware of of what should we do with it now as a developer.
So maybe you could explain that a bit more.

Author 1 1:09:51
Yeah. OK.
Concerning the use case, of course, here we presented the very simple use case about irrigation plant.
But of course we we plan to add additional use case in and of course evaluate the complexity of the the specification that may rise as you suggest, but.
Also.
Using our tool actually.
We have the possibility to specify different specification file, so to have a modular modular implementation of the older rules that we specified because.
The underpinning technology that we are using actually is X base.
That is, let's say it's related to X, but allows us to import also other specification files.
So in that regard, I agree that in the in the demo we we we show a very long specification, but we can of course we can rely on this capabilities to.
Different specification files or the user can.
Can maybe define the first of the the ports and then on another side. But of course this this should be guided in some way with the some model of the complexion tool, or at least.
Additional additional guidelines on how to use the DSL.
Coming back to the last questions about the discovery, of course this is a preliminary set and we.

We show that the output as a Jason file, but in the end we we can have extended. We can.
We can of course extend the discovery component.
By by adding, for instance, some automated metric or at least a complete report on what what's happening on the system.
I mean the the The thing is that of course the simulation environment.
Is deducted from the DSL environment.
So for instance, we cannot report, for instance, the complete log of the simulation, even though even though according to the simulation platform that we also we we can have also detailed report on that.
So of course we can.
The discovery component can be can be extended in that way.

Author 4 1:12:27
But maybe let me just add one thing about the motivation.

Author 1 1:12:30
Yeah.

Author 4 1:12:32
Let's say behind the the the work we started from the failure propagation analysis. So studying our failures propagating IoT systems and this is usually done starting from rules that specifies how failures may propagate among individual components. And then there are tools doing the analysis to study how the.
Failures may propagate may propagate across the entire.
System the issues that if the rules about the individual components are inaccurate, then the whole analysis may be inaccurate.
So the idea of the tool is to provide an experimental, let's say framework, where you could run the tools and you could run the components, inject the failures in input and study how this failure propagates for the individual components so that you could infer these the rules about.
The individual component, let's say on the field by running the component. Then you could explore.
This information to study.
How these failures may propagate across the entire system?
So this is this is how we are somehow exploiting the information that we discover about

this, let's say one step propagation related to the individual components that are executed through this framework.

Author 2 1:13:46

And if I can comment on also on the, on the initial question from Judy today related to the injection pattern. So I would like to verify the fact that with this notion of component that we have in the law in the, in the language we did with the.
Component we refer to even a group of real components, so we group all of them and the input. Now that we specify or that we consider is not a necessary one specific input.
Top very specific components, so we'd like to analyze very sensitive components that are in connected among them and somehow and we will look at the injection of the initial compon by connecting to one input of the neural, one of the component and look at the output of comple.
Different components, right?
And I can run this analysis and I can, you know.
Sub I can put a process right? So if said there is something wrong?
I can maybe focus on the different set of components by reducing for instance the scope of my of my analysis.
I would like I wanted to to to clarify also this point. If this was not clear.

Author 1 1:15:04

OK.
I guess we can go to **omissis**.

Participant 5 1:15:10

Yes. So maybe if I can add something that was not.
Like besides what other people already said.
So it's maybe a bit more related to maybe our previous work about the failure types and.
The way the types are models through the injection rules, if I understand it correctly.
So my question is whether these failure types were somehow instructed by.
That happen in the field.
And.
Whether.
The language also, I don't know if it makes sense to do that for whether it makes sense to combine these these failure types together, and whether the language allows to do that.

Author 1 1:16:09
Uh yes, this is allowed because, as also **omissis**was mentioned before, IA developer can have their own definition of values after.

Participant 5 1:16:16
Mm hmm.
Huh.

Author 1 1:16:21
And of course.
We provide in the standard library, I mean the built in function that we developed.
We provide also atomic function like like I showed in the demo for like changing a value or maybe changing just the time.
To and and these kind of atomic injection that can be combined in a single function that the user can of course do during the specification.
And.
This this failure, of course, comes to.
Previous work in the literature and we of course they are they they've been observed in simulation environment or real real environment limited of course in the assuming that the system of course is.
Based on time series, so it's related of course on the IS mostly related of course on the time that the value comes to the system.
But yeah, the the.
Can the motivational on top of the for the for our tool is to provide a certain degree of freedom from the developer perspective so they can they can implement their own failure and maybe not.
Why not combine combine for instance a nearly failure with a lead failure to to to create.
To create another kind of failure that.
Can be can be.
Maybe most most suitable for the type of the use case that I mean in IoT domain the user wants to support.
So definitely the yeah, this is possible.

Participant 5 1:18:14
Yeah. So as a as a follow up.

So when, for instance, you combine multiple failures or you define your own.
The.
When you compare the expected.
So like this type of failure is injected at once.
Not like in multiple steps.
So you have at once. OK, so at the end at the end you need to compare like the the expected time series.

Author 1 1:18:35
No. Yeah.
Yeah, except.

Participant 5 1:18:44
This with the one and I was wondering whether.
It makes sense to.
To be able to distinguish between the two failures.
So whether the analysis has no way to do that.

Author 3 1:19:01
Yes it is possible.
So if I can add some things.

Participant 5 1:19:03
OK.

Author 3 1:19:04
So here the was saying there are two steps.
The first one we are defining the injection and the second one we are defining the discovery part in the discovery part. We can reuse a set of atomic function. For instance adding a specific value.

Participant 5 1:19:13
Yeah.

Author 3 1:19:20
What specific so checking if a specific value, what specific position?

Must exchange it or not or we can reuse this one that is showing in the slide 12.
Where we are complex failure discovery and moreover those one are some parameter for reasons. If you would like to check failure or some things like that we have to define an epsilon time. We can define some parameter for epsilon time to check if it is within node or.

Participant 5 1:19:38
Mm hmm.

Author 3 1:19:52
Not these these sets for over these complex failures, the atomic ones can be combined and define more complex.
Discovery, discovery operations.
Shoes.

Participant 5 1:20:06
OK.
Yeah, so basically this.
This failure types.
Are.
Let you know. Wait, they don't.
They don't interfere, so that that's why it is possible to distinguish between them.

Author 3 1:20:22
Yeah. Excellent. Yeah. Yeah, yeah.

Participant 5 1:20:23
OK.
OK.

Author 1 1:20:37
This is AI think that we can extend for future version of the tool.

Participant 5 1:20:44
OK.
Thank you.

Author 1 1:20:46

Thanks to you, I think Pablo.

Participant 6 1:20:50

Yes, thanks.

Well I have.

Some.

Well, a suggestion I think is related to.

Well, I think it's very usefulness if I.

Is very useful.

Maybe the DSL well, I think.

This is still needs like more conciseness.

But, but I think it's.

Question of.

Maybe I'm wrong. OK, but I suppose it's as you are injecting. Also this Java code.

Well, I think it's it's it's good also.

I was wondering if you can inject.

As you said.

Jason.

Code also in the DSL if I'm not wrong.

Author 1 1:22:04

No, no, this is for let's say the input files.

Author 3 1:22:08

Input output.

Author 1 1:22:09

Nothing input output file yeah.

Participant 6 1:22:09

Ah, yes, OK.

OK. But maybe you can also inject.

I don't know any other DSL in your DSL, if I may ask this.

Author 1 1:22:25

This is something that we have to check actually, but I guess maybe you you can import different DSL file.

Participant 6 1:22:28

Yes.

Author 3 1:22:29

Who?

Participant 6 1:22:35

OK.

Well, that's also a good point that I can check in my prayers, OK.

So well, just another just, well, I think it's very, very interesting. And we are also in the midsto group.

We have also a research line very similar to this one.

And.

Well, I would say.

What about adding a graphical layer for representing the the testing process over the IoT environment?

Do you think it could be?

Much, much complicated or much complex.

To include these well, I think it could be like for a presentation or something or a demo to to show the the testing process.

In in execution under execution.

I think it's related also to what?

Maybe Participant 4was saying.

About the reports or the results of the testing process, well, maybe.

Yeah.

Author 1 1:24:04

Sure, sure. This is very good feedback.

Participant 6 1:24:08

Mm hmm.

Author 1 1:24:08
Yeah, for this for this initial demo, let's say we stick with. When I mean state-of-the-art platform because we want to do stress depart of the Texan specification.

Participant 6 1:24:13
Yes.
Yes.

Author 1 1:24:22
But of course graphical abstraction like for instance with the series based.
Tools or maybe? Yeah. Integration.
Another gift set could be indeed use a folder to monitor the process.

Participant 6 1:24:39
Mm hmm.

Author 2 1:24:40
If I can comment please on power on this comment, Pablo.

Participant 6 1:24:43
Yes.

Author 2 1:24:46
So. So we currently, now we we have seen how you you can simulate your system by using which is already graphical non graphic based.
Approach. So here what you could represent graphically is the set of components, right?
But concerning the other parts about the you know the injections, there are very non textual specific artifacts.
Of course, I mean, since you are on your math, I mean developing a serious base the editor to graphically represent the components that you but this would.

Participant 6 1:25:13
Yes.

Author 2 1:25:23

Yeah, it's just for because it is possible to do, but concerning this ability we have to understand if it is actually useful in addition to already graphical base representation of your system that you have already inputels, right?

Participant 6 1:25:30
Yes.
Yes.

Author 2 1:25:38
But OK, we take this comment, of course.
Even though, I mean, I think that this would be kind of duplication no because and also because most of the of the injections and all this kind of analysis that we do are essentially textile based.
I mean you want to.
You have to look at actually time series that you graph that you textually to to represent.

Participant 6 1:26:05
OK. Thanks. Yes.

Author 3 1:26:06
Just to to complete the answer with just to complete the answer with the the motivation of using Java code inside the rdsl is the fact that it's extend ext is well supported by the extended section and we reuse it.
This is syntax syntax and the semantics of extend extension.

Participant 6 1:26:31
OK.
Yes, it's completely.
Like found.
Found it, I think it's.
It's a good a good approach.
Well, I I think I don't have many more comments.
Maybe my doubt was related to my.
Low.
Skills on this?
Platforms to simulate like protose and all these.

Related works.
Well, I think maybe another.
Well, just a question because as you know, I create a DSL for model mutation, yes, so.
These operators that you are well, the failures you introduce are based on your own expertise.
Or are they devised by?
Different.
Well, just if you can share some thoughts about this.
Thank you.

Author 3 1:27:51
OK.
So we we were driven by the, the, the the needs of implementing some complex.
Complex discovery and mutation and mutational rules. So we, we.
Function but for sure for defining these atomic function we look at around the literature.
For sure, we also look at the the work that your I guess that you are mentioning Woden.
And we we were also inspiring by the the DEMUTATION rules that in order.

Participant 6 1:28:26
OK.

Author 2 1:28:27
Even though, yeah, we got expiration from that, we know very well.

Author 3 1:28:30
Yeah.

Author 2 1:28:31
I mean your your approach, **omissis** and we got inspired by you know, the way of thinking, even though here, I mean this is also connected to the previous comment from your side.
So here we don't want to.
I mean we we don't want to to change our model.
So we don't want to mutate our model, right?

Participant 6 1:28:48
Yes.

Author 2 1:28:50
So the the, the the architecture of the system would be is essentially the same.
So once we define another, the set of components.
And we are not mutating the architecture. We are not adding or changing, we are just I mean triggering not execution of this component by changing the inputs.

Participant 6 1:29:00
Yes.

Author 2 1:29:08
Now This is why is not appropriate for for this this case, even though of course Conceptally is essentially is the same. Now it's a different level of of. Intervene over changing.

Participant 6 1:29:09
Yes.
Yes.
Yes, yes.
Thank you.
Yes, I I I was thinking about when I.
Listen to you. This presentation I was thinking about. Well, how can we?
Apply world to this, but it's not as as close to the the way you are.
All these mutation operators, as you are saying are are different.
In in this kind of application, yes, yes.
I suppose.
Well.
There are some other approaches that are are.
Closer to to this way of applying the operators, but it's very nice to see this work is is also.
Benefiting from?
For for the the 1:00 we, we we are developing since seven years ago.
Yes.
OK.
Well, thank you.

I I think I have no more comments.
Thank you, Claudio and Judy and David.

Author 2 1:30:49
Thank you.

Author 1 1:30:51
OK.
So I don't know if there is other further comments.

Author 2 1:31:04
OK, so I think that, yeah, if there are no other comments, I would like to thank you all for.
I mean for rapping us and for I mean providing you know feedback very valuable feedback and yeah.
So see you around next time and.

Author 1 1:31:26
OK.

Author 2 1:31:26
And.

Participant 1 1:31:26
Repl.

Author 1 1:31:27
Thank you again.

Participant 6 1:31:29
For the usability.
Experiment of the tool, yes.

Author 2 1:31:33
Yeah.

Author 1 trascrizione arrestata